

Numerical Simulation of Phase Transformation Kinetics and Microstructure Evolution in Metals using GNU Octave.

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Abstract: Phase transformation and microstructure evolution play a fundamental role in determining the mechanical and thermal properties of metallic materials. Understanding these transformations is important in materials engineering, particularly in processes such as heat treatment, alloy development, and thermal processing. In this project, a numerical simulation model was developed using GNU Octave to study phase transformation kinetics and microstructure evolution in metals.

The simulation was based on the Johnson-Mehl-Avrami-Kolmogorov equation to describe transformation kinetics under different temperature conditions. Temperature-dependent transformation behavior was analyzed using Arrhenius-type relations, and an approximate Time-Temperature-Transformation diagram was generated to visualize transformation start and finish conditions. In addition, a two-dimensional grid-based model was implemented to simulate microstructure evolution and phase growth during transformation.

The computational model successfully generated phase transformation curves, cooling curves, transformation rate analysis, and microstructure evolution patterns. Numerical parameters such as transformation times (t_{10} , t_{50} , t_{90}), cooling rates, and transformed phase fractions were also evaluated. The results demonstrated the strong influence of temperature on transformation kinetics and microstructural development.

This project highlights the usefulness of computational modelling in materials science and demonstrates how numerical simulation tools can be applied to study metallurgical phenomena. The work also provides a foundation for further research in computational materials science, phase-field modelling, and advanced microstructure simulations.

Keywords: Phase Transformation Kinetics, Microstructure Evolution, Johnson-Mehl-Avrami-Kolmogorov Model, Time-Temperature-Transformation Diagram, Computational Metallurgy, Numerical Simulation, GNU Octave.

1. Introduction

Phase transformation and microstructure evolution are among the most important phenomena in materials science and metallurgical engineering. The mechanical, thermal, and structural properties of metallic materials are strongly influenced by their internal microstructure and transformation behavior during thermal processing. Processes such as heat treatment, annealing, quenching, and alloy solidification involve complex phase transformations that directly affect the performance and reliability of engineering materials.

Understanding transformation kinetics is essential for predicting the behavior of metals under different thermal conditions. The transformation of one phase into another is governed by nucleation, grain growth, and atomic diffusion mechanisms. These processes are commonly studied using mathematical and computational models to analyze the relationship between temperature, time, and phase evolution. Computational simulations provide an efficient way to visualize and study these transformations without performing expensive experimental investigations.

In recent years, computational materials science has become increasingly important in the development and optimization of advanced engineering materials. Numerical simulation techniques are widely used to study phase transformation behavior, microstructure evolution, and thermal processing in metallic systems. Such simulations help researchers understand transformation mechanisms, estimate transformation rates, and predict material behavior under different operating conditions.

In this project, a numerical simulation of phase transformation kinetics and microstructure evolution in metals was developed using GNU Octave. The simulation was based on the Johnson-Mehl-Avrami-Kolmogorov equation to model transformation kinetics under varying temperature conditions. In addition, a two-dimensional grid-based microstructure model was implemented to visualize phase evolution and grain growth behavior. The project also included the generation of an approximate Time-Temperature-Transformation diagram and the analysis of cooling behavior and transformation rates.

The primary objective of this work is to demonstrate how computational modeling and numerical simulation techniques can be applied to study metallurgical phenomena and microstructural evolution in materials systems. The project further aims to provide a foundational understanding of computational approaches used in modern materials science research.

2. Literature Review

Phase transformation and microstructure evolution have been widely studied in the field of materials science because of their significant influence on the mechanical and thermal properties of metallic materials. The relationship between processing conditions, microstructural development, and material performance has been an important area of research in metallurgy and engineering materials for many decades. Researchers have developed various theoretical and computational approaches to understand transformation kinetics, grain growth, and diffusion-controlled processes in metals and alloys.

One of the most widely used models for studying phase transformation kinetics is the Johnson-Mehl-Avrami-Kolmogorov equation. This model describes the transformation of phases as a function of time and temperature and is commonly applied in the analysis of crystallization, solid-state transformation, and heat treatment processes. The JMAK model provides a mathematical relationship between transformed phase fraction and transformation time, making it useful for predicting the kinetics of nucleation and grain growth in metallic systems [1].

M. C. Weinberg, D.P. Birnie and V. A. Shneidman et.al., 1997 have further expanded the JMAK theory to improve its applicability to real material systems. Weinberg et al. discussed the limitations of the classical JMAK equation under conditions such as anisotropic particle growth, non-uniform nucleation, and finite-size effects. Their work demonstrated that practical phase transformations often deviate from ideal assumptions and require more advanced modeling approaches [2].

The evolution of microstructure during phase transformation has also been studied using computational and statistical approaches. E. Pineda and D. Crespo et.al., 1999 developed a statistical model for analyzing grain-size distribution and microstructure evolution during nucleation and growth kinetics. Their study highlighted the importance of grain impingement and growth history in determining final microstructural characteristics [3].

With the advancement of computational techniques, numerical simulations and Monte Carlo methods have become increasingly important in the study of transformation kinetics. B.J. Kooi et.al., 2004 investigated anisotropic growth behavior using Monte Carlo simulations and demonstrated the limitations of classical JMAK assumptions in systems with directional grain growth. These studies emphasized the importance of computational approaches in understanding realistic transformation behavior in materials systems [4].

Recent developments in computational metallurgy have focused on combining numerical simulation with microstructure prediction and grain-size evolution analysis. Zimmer et al., 2017 revisited the classical JMAK framework and applied it to ferrite grain evolution in steels using mathematical and computational modeling approaches. Their work demonstrated the significance of computational methods in predicting material behavior and microstructural development [5].

The present project builds upon these theoretical and computational approaches by combining phase transformation kinetics, microstructure evolution, cooling behavior, and numerical simulation into a single computational framework. The study demonstrates the practical application of computational methods in understanding and visualizing transformation behavior in metallic materials.

3. Methodology

The present work was carried out using a computational and numerical simulation approach to study phase transformation kinetics and microstructure evolution in metallic materials. The entire simulation framework was developed independently using GNU Octave, which was selected because of its strong numerical computing and visualization capabilities. The methodology of the project consisted of mathematical modeling, numerical implementation, graphical visualization, and quantitative analysis of transformation behavior under different thermal conditions.

The first stage of the work involved the development of a phase transformation kinetics model based on the Johnson-Mehl-Avrami-Kolmogorov equation. This model was used to calculate the transformed phase fraction as a function of time and temperature. Temperature-dependent transformation behavior was incorporated using an Arrhenius-type relation for the transformation rate constant. Different temperature conditions were considered in order to analyze the influence of thermal variation on transformation kinetics and phase evolution.

A numerical simulation was then implemented to generate phase transformation curves and evaluate important transformation parameters such as transformation start time, transformation completion time, and transformation rate. The transformed phase fraction was calculated at multiple time intervals to study the progression of phase evolution during the simulation.

To simulate microstructure evolution, a two-dimensional grid-based model was developed. In this approach, each grid element represented a small region of the material microstructure. Initial nuclei were randomly generated within the computational grid, and grain growth was simulated using probabilistic neighborhood-based rules. The transformation process was iteratively updated over multiple simulation steps, allowing visualization of grain growth and transformed regions within the material.

An approximate Time-Temperature-Transformation diagram was also generated using transformation start and finish conditions calculated from the numerical model. This diagram was used to study the effect of temperature on transformation kinetics and phase stability.

In addition to phase transformation analysis, a cooling curve model was implemented to study the thermal behavior of the material during cooling. Cooling rates were calculated numerically to examine their influence on transformation behavior and microstructure development.

The results obtained from the simulations were visualized using graphs, transformation curves, TTT diagrams, and microstructure evolution images. Numerical parameters including transformation fractions, transformation rates, and cooling rates were extracted and analyzed to evaluate the overall behavior of the simulated material system.

4. Implementation

The computational implementation of the project was carried out using GNU Octave. The program was designed in a modular and structured manner to simulate phase transformation kinetics, microstructure evolution, transformation analysis, and thermal behavior in metallic materials. Numerical calculations, matrix operations, and graphical visualization tools available in GNU Octave were extensively used throughout the project.

The implementation process began with the initialization of material and simulation parameters such as temperature range, simulation time, activation energy, gas constant, transformation exponent, and transformation rate constants. These parameters were used to establish the computational framework for the phase transformation model.

The Johnson-Mehl-Avrami-Kolmogorov equation was implemented numerically to calculate the transformed phase fraction at different time intervals and temperature conditions. Iterative loops were used to compute phase evolution over time, and the resulting transformation curves were plotted to analyze transformation kinetics. Numerical calculations were also performed to determine important transformation parameters such as t_{10} , t_{50} , and t_{90} values, representing the times required for 10%, 50%, and 90% transformation respectively.

To study the effect of temperature on transformation behavior, temperature-dependent transformation constants were calculated using an Arrhenius-type relation. Multiple transformation curves corresponding to different temperatures were generated and compared to analyze the influence of thermal conditions on transformation rates and phase evolution.

The approximate Time-Temperature-Transformation diagram was generated by calculating transformation start and finish conditions over a selected temperature range. Logarithmic plotting functions were used to visualize transformation boundaries and thermal behavior during the simulation.

For microstructure evolution analysis, a two-dimensional matrix-based grid model was implemented. Each matrix element represented a localized region of the material microstructure. Randomly distributed nuclei were introduced into the computational grid, and grain growth behavior was simulated using neighborhood-based probabilistic transformation rules. The microstructure was updated iteratively over multiple simulation steps, allowing visualization of transformed and untransformed regions during phase evolution.

A cooling curve simulation was also implemented using an exponential cooling relation. Temperature variation with time was calculated numerically, and cooling rates were extracted from the generated data. The cooling analysis was used to examine the influence of thermal conditions on transformation behavior and microstructure development.

The graphical outputs generated during the implementation included phase transformation curves, cooling curves, TTT diagrams, and microstructure evolution images. These outputs were saved in high-quality image formats for further analysis and documentation in the research report.

5. Report

The numerical simulation successfully modeled phase transformation kinetics and microstructure evolution in metals under different thermal conditions using GNU Octave. The computational model generated phase transformation curves, an approximate Time-Temperature-Transformation diagram, microstructure evolution patterns, and cooling behavior analysis. Both qualitative and quantitative results were obtained from the simulation.

The phase transformation curves generated from the Johnson-Mehl-Avrami-Kolmogorov equation demonstrated the influence of temperature on transformation kinetics. It was observed that higher temperatures resulted in faster phase transformation behavior due to an increase in transformation rate constants. The transformation curves showed the characteristic sigmoidal behavior commonly associated with nucleation and growth-controlled transformations in metallic systems.

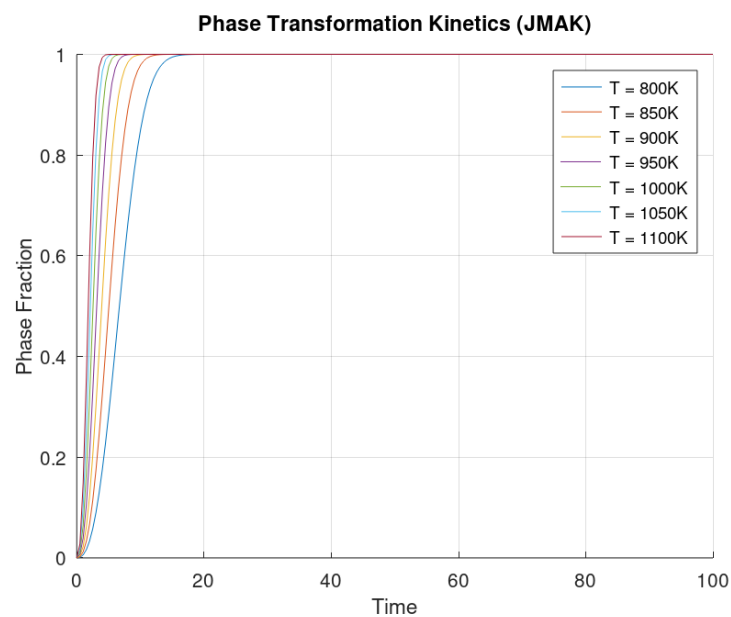


Figure-01: Phase Transformation Curves at Different Temperatures

The approximate TTT diagram generated from the simulation illustrated the relationship between temperature and transformation time. The diagram indicated that transformation start and finish times varied significantly with temperature. The transformation process occurred more rapidly within specific temperature ranges, demonstrating the strong dependence of phase kinetics on thermal conditions.

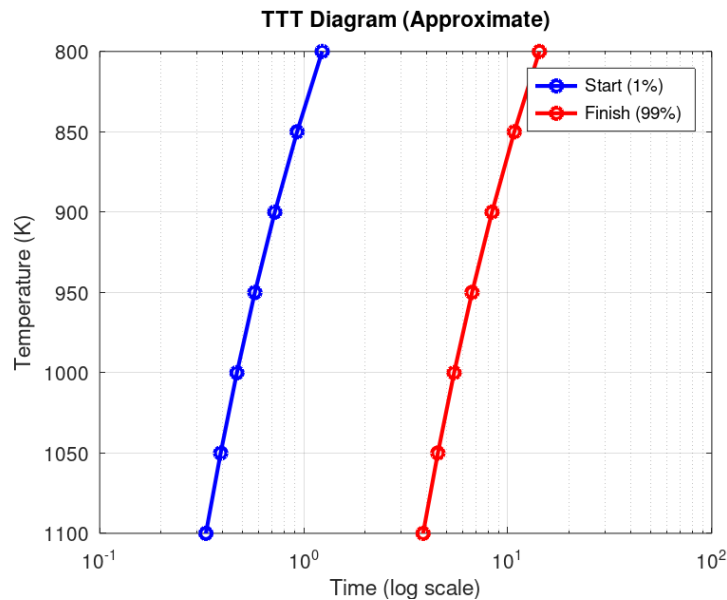


Figure-02: Approximate Time-Temperature-Transformation(TTT) Diagram

A two-dimensional grid-based microstructure model was implemented to visualize phase growth and grain evolution during transformation. The simulation showed progressive phase expansion from initial nucleation sites, eventually leading to a larger transformed region within the material domain. The generated microstructure evolution images qualitatively represented grain growth and phase propagation behavior during thermal processing.

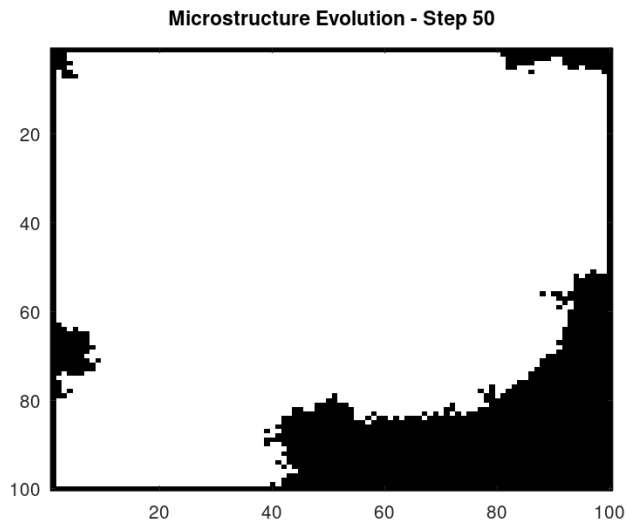


Figure-03: Simulated Microstructure Evolution During Phase Transformation

The cooling curve simulation demonstrated the decrease in temperature with time under exponential cooling conditions. The calculated cooling rate values indicated that thermal history strongly affects phase transformation behavior and transformation completion time. Faster cooling conditions reduced the available transformation time, whereas slower cooling allowed greater phase development and grain growth.

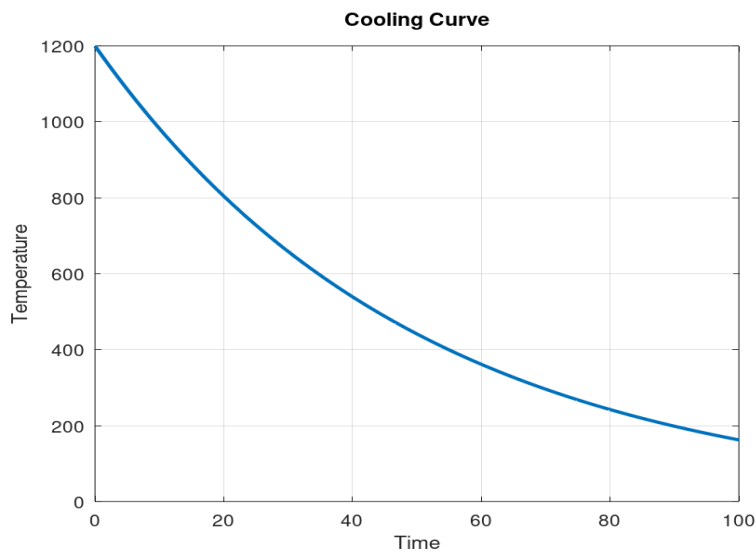


Figure-04: Cooling Curve Generated from Numerical Simulation

Quantitative analysis was also performed using the simulated transformation data. Transformation parameters such as t_{10} , t_{50} , and t_{90} were calculated to determine the time required for 10%, 50%, and 90% phase transformation, respectively. In addition, transformation rate, final transformed phase fraction, and average cooling rate were evaluated for different thermal conditions.

Table-01: Calculated Transformation Parameters From Numerical Simulation

Temperature(K)	$t_{10}(s)$	$t_{50}(s)$	$t_{90}(s)$	Maximum Transformation Rate(/s)	Final Phase Transformation
800	3.52	7.04	11.06	0.1300	1.00
850	2.51	5.53	8.54	0.1726	1.00
900	2.01	4.02	6.53	0.2215	1.00
950	1.51	3.52	5.53	0.2749	1.00
1000	1.51	3.02	4.52	0.3360	1.00
1050	1.01	2.51	3.52	0.3945	1.00
1100	1.01	2.01	3.02	0.4764	1.00

The numerical results demonstrated that the transformation rate increased with temperature, while the required transformation time decreased. The microstructure fraction analysis further indicated progressive phase evolution with increasing simulation time. Overall, the obtained results were consistent with fundamental principles of phase transformation kinetics and computational metallurgy.

6.Quantitative Analysis

The numerical simulation provided several important quantitative parameters for evaluating phase transformation kinetics under different thermal conditions. Transformation times corresponding to 10%, 50%, and 90% transformation fractions (t_{10} , t_{50} , and t_{90}) were calculated for temperatures ranging from 800 K to 1100 K. In addition, the maximum transformation rate and final transformed phase fraction were determined to assess the influence of temperature on transformation behavior.

The results indicate a strong dependence of transformation kinetics on temperature. As the transformation temperature increased from 800 K to 1100 K, the characteristic transformation times decreased significantly. The value of t_{10} decreased from 3.52 s at 800 K to 1.01 s at 1100 K, while t_{50} decreased from 7.04 s to 2.01 s over the same temperature range. Similarly, the t_{90} value decreased from 11.06 s to 3.02 s, indicating that the overall transformation process became progressively faster at higher temperatures.

A substantial increase in the maximum transformation rate was also observed. The maximum rate increased from 0.1300 s^{-1} at 800 K to 0.4764 s^{-1} at 1100 K. This trend suggests enhanced nucleation and growth kinetics with increasing temperature. The increase in thermal energy promotes atomic mobility and accelerates phase growth, resulting in shorter transformation times and higher transformation rates.

The final transformed phase fraction reached a value of 1.00 for all investigated temperatures. This indicates that complete phase transformation was achieved under the selected simulation conditions regardless of temperature. However, the time required to achieve complete transformation varied significantly, demonstrating the critical role of temperature in controlling transformation kinetics.

The quantitative results are consistent with classical metallurgical theory, which predicts that increasing temperature accelerates diffusion-controlled phase transformations. The observed reduction in transformation times and increase in transformation rates confirm the validity of the computational model developed in this study. These findings also demonstrate the capability of numerical simulation techniques to predict phase evolution and transformation behavior in metallic materials.

Overall, the quantitative analysis highlights the effectiveness of the developed GNU Octave model in describing phase transformation kinetics and provides valuable insights into the relationship between temperature, transformation rate, and microstructural evolution.

7.Discussion

The results obtained from the numerical simulation provide valuable insights into the kinetics of phase transformation and microstructure evolution in metallic materials. The generated transformation curves exhibited the characteristic sigmoidal behavior commonly associated with nucleation-and-growth-controlled transformations. This behavior is consistent with classical phase transformation theory and demonstrates the suitability of the Johnson-Mehl-Avrami-Kolmogorov (JMAK) model for describing transformation kinetics under varying thermal conditions.

A significant observation from the simulation was the strong influence of temperature on transformation behavior. As the transformation temperature increased from 800 K to 1100 K, the transformation times required to achieve 10%, 50%, and 90% phase conversion decreased considerably. This reduction in transformation time can be attributed to the increased atomic mobility at elevated temperatures, which enhances nucleation and growth processes. Consequently, higher temperatures promoted faster transformation kinetics and accelerated phase development.

The generated Time-Temperature-Transformation (TTT) diagram successfully illustrated the relationship between transformation time and temperature. The diagram demonstrated that specific temperature ranges are more favourable for rapid transformation, while lower temperatures require longer transformation periods. Although the model represents a simplified approximation of actual metallurgical systems, the generated trends are physically meaningful and consistent with theoretical applications.

The microstructure evolution simulation provided a qualitative representation of grain growth and phase propagation during transformation. The transformation initiated from randomly distributed nucleation sites and gradually expanded throughout the computational domain. As the simulation progressed, transformed regions grew and merged with neighboring regions, resulting in a continuously evolving microstructure. This behavior reflects the fundamental

mechanisms of nucleation and grain growth observed in real metallic systems during heat treatment processes.

The cooling curve analysis highlighted the importance of thermal history in determining transformation behavior. Variations in cooling conditions can significantly affect the available time for nucleation and growth, thereby influencing the final microstructure and material properties. The simulation demonstrated how cooling rate can alter transformation progression and ultimately affect phase evolution.

Despite the successful implementation of the model, certain limitations should be acknowledged. The simulation was based on simplified assumptions and did not incorporate complex phenomena such as anisotropic grain growth, multi-phase interactions, composition-dependent diffusion, or stress-assisted transformations. Furthermore, the generated TTT diagram and microstructure evolution model represent idealized conditions rather than specific industrial alloy systems. Nevertheless, the model effectively demonstrates the fundamental principles governing phase transformation kinetics and microstructural development.

Overall, the results obtained in this study confirm that numerical simulation is a powerful tool for investigating metallurgical phenomena. The developed GNU Octave model successfully captured the essential features of phase transformation kinetics and microstructure evolution, providing a computational framework for understanding transformation behavior in metallic materials. The study also highlights the potential of computational approaches in supporting materials design, process optimization, and advanced materials research.

8. Conclusion

This study successfully demonstrated the numerical simulation of phase transformation kinetics and microstructure evolution in metals using GNU Octave. The developed computational model effectively represented the transformation process through the Johnson-Mehl-Avrami-Kolmogorov (JMAK) equation and provided a simplified framework for analyzing nucleation, phase growth, and temperature-dependent transformation behavior.

The simulation generated phase transformation curves, an approximate Time-Temperature-Transformation (TTT) diagram, cooling profiles, and two-dimensional microstructure

evolution patterns. In addition, quantitative parameters such as t_{10} , t_{50} , t_{90} transformation times, maximum transformation rate, cooling rate, and final transformed phase fraction were successfully calculated and analyzed. The obtained results showed that increasing temperature significantly accelerated transformation kinetics by reducing transformation times and increasing the maximum transformation rate, while complete phase transformation was achieved under all simulated conditions.

The microstructure evolution model qualitatively illustrated the growth of transformed regions from randomly distributed nucleation sites, providing a visual understanding of grain development during thermal processing. Although the model employed several simplifying assumptions and does not represent the full complexity of industrial alloys systems, it successfully captured the fundamental characteristics of diffusion-controlled phase transformations and demonstrated the effectiveness of numerical methods in studying metallurgical phenomena.

From an educational and research perspective, this project highlights the value of computational simulation as a cost-effective tool for investigating material behavior and understanding phase transformation mechanisms. The developed model serves as a useful foundation for further studies involving more advanced approaches such as phase-field modeling, CALPHAD-based simulations, finite element analysis, and data-driven materials design.

Overall, the project achieved its intended objectives and demonstrates how numerical techniques can be applied to model important materials science phenomena. The findings emphasize the growing role of computational methods in modern metallurgy and provide a basis for future work aimed at improving the accuracy and applicability of phase transformation simulations in engineering materials.

9.Future Scope

The present work provides a simplified computational framework for studying phase transformation kinetics and microstructure evolution in metallic materials. In future studies, the model can be extended by incorporating real experimental material data and thermodynamic parameters to improve the accuracy of transformation predictions.

The implementation of composition-dependent diffusion, multi-phase transformations, and alloy-specific properties would make the simulation more representative of practical engineering applications.

Further improvements may include the integration of advanced computational approaches such as phase-field modelling, finite element analysis, and CALPHAD-based thermodynamic calculations. These techniques would enable more realistic simulations of grain growth, solidification behavior, and complex microstructural evolution under processing conditions.

Additionally, the developed model can be enhanced by incorporating machine learning and artificial intelligence techniques for predicting phase transformations and optimizing heat-treatment processes. Such developments could contribute to modern computational materials science and support the design of advanced metallic material with improved performance and reliability.

10. References

- [1]. M. Fanfoni and M. Tomellini, “The Johnson-Mehl-Avrami-Kolmogorov model: A brief review,” *Il Nuovo Cimento D*, 1998.
- [2]. M. C. Weinberg, D. P. Birnie, and V. A. Shneidaman, “Crystallization kinetics and the JMAK equation,” *Journal of Non-Crystalline Solids*, 1997.
- [3]. E. Pineda and D. Crespo, “Microstructure development in Kolmogorov, Johnson-Mehl, and Avrami nucleation and growth kinetics,” *Physical Review B*, 1998.
- [4]. B. J. Kooi, “Monte Carlo simulations of phase transformations caused by nucleation and subsequent anisotropic growth,” *Physical Review B*, 2004.
- [5]. J. Zimmer et al., “A revisited Johnson-Mehl-Avrami-Kolmogorov model and the evolution of grain-size distributions in steel,” *IMA Journal of Applied Mathematics*, 2017.

